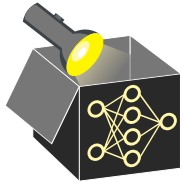
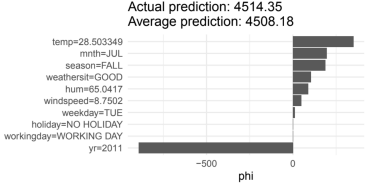
# Interpretable Machine Learning



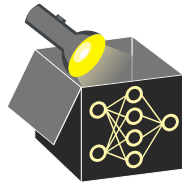
## Shapley
## Shapley Values for Local Explanations



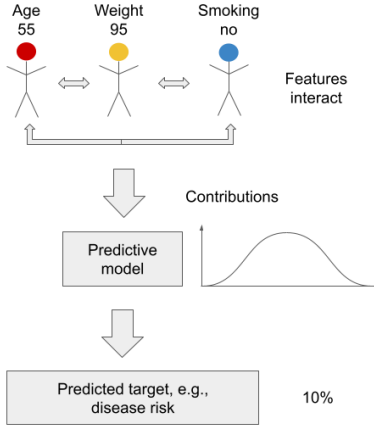

**Learning goals**

- See model predictions as a cooperative game
- Transfer the Shapley value concept from game theory to machine learning

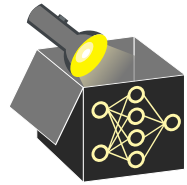
# FROM GAME THEORY TO MACHINE LEARNING



- Model prediction depends on feature interactions for a specific observation
- **Goal:** Decompose prediction into **individual feature contributions**
- **Idea:** Treat features as players jointly producing a prediction
- How to fairly assign credit to features? $\Rightarrow$ Shapley values



# FROM GAME THEORY TO MACHINE LEARNING

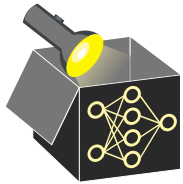


- **Game:** Predict $\hat{f}(x_1, x_2, \ldots, x_p)$ for a single observation $\boldsymbol{x}$
- **Players:** Features $x_j, j \in \{1, \ldots, p\}$, cooperate to produce a prediction
- **Value function:** Defines payout of coalition $S \subseteq P$ for observation $\boldsymbol{x}$ by

$$v(S) = \hat{f}_S(\boldsymbol{x}_S) - \hat{f}_\emptyset, \text{ where}$$

  - $\hat{f}_S : \mathcal{X}_S \mapsto \mathcal{Y}$ is the PD function $\hat{f}_S(\boldsymbol{x}_S) := \int \hat{f}(\boldsymbol{x}_S, X_{-S})\, d\mathbb{P}_{X_{-S}}$
    $\rightsquigarrow$ "Removes" features in $-S$ by marginalizing, keeping $\hat{f}$ fixed
  - Mean prediction $\hat{f}_\emptyset := \mathbb{E}_{\boldsymbol{x}}(\hat{f}(\boldsymbol{x}))$ is subtracted to ensure $v(\emptyset) = 0$
- **Goal:** Distribute total payout $v(P) = \hat{f}(\boldsymbol{x}) - \hat{f}_\emptyset$ fairly among features

# FROM GAME THEORY TO MACHINE LEARNING



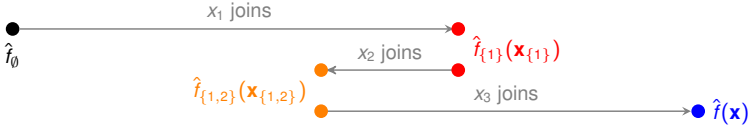
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- **Goal:** Distribute total payout $v(P) = \hat{f}(\boldsymbol{x}) - \hat{f}_\emptyset$ fairly among features
- **Marginal contribution of feature $j$ joining coalition $S$** ($\hat{f}_\emptyset$ cancels):

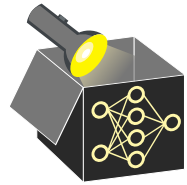
$$\Delta(j, S) = v(S \cup \{j\}) - v(S) = \hat{f}_{S\cup\{j\}}(\boldsymbol{x}_{S\cup\{j\}}) - \hat{f}_S(\boldsymbol{x}_S)$$

- **Example (3 features):** Feature contributions for joining order $x_1 \to x_2 \to x_3$ toward total payout $v(P) = \hat{f}(\boldsymbol{x}) - \hat{f}_\emptyset$, each step reflects a marginal contribution



# SHAPLEY VALUE - DEFINITION

▸ "Shapley" 1953 ▸ "Strumbelj et al." 2014



**Order definition:** Shapley value $\phi_j(\boldsymbol{x})$ quantifies contribution of $x_j$ via

$$\phi_j(\boldsymbol{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \underbrace{\hat{f}_{S_j^\tau \cup \{j\}}(\boldsymbol{x}_{S_j^\tau \cup \{j\}}) - \hat{f}_{S_j^\tau}(\boldsymbol{x}_{S_j^\tau})}_{\Delta(j, S_j^\tau) \text{ marginal contribution of feature } j}$$

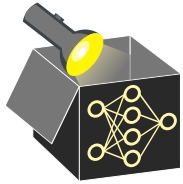
- **Interpretation:** $\phi_j(\boldsymbol{x})$ quantifies how much feature $x_j$ contributes to the difference between $\hat{f}(\boldsymbol{x})$ and the mean prediction $\hat{f}_\emptyset$
  $\rightsquigarrow$ Marginal contributions and Shapley values can be negative
- **Exact computation of $\phi_j$:** Using PD function $\hat{f}_S(\boldsymbol{x}_S) = \frac{1}{n} \sum_{i=1}^n \hat{f}(\boldsymbol{x}_S, \boldsymbol{x}_{-S}^{(i)})$ yields

$$\phi_j(\boldsymbol{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \frac{1}{n} \sum_{i=1}^{n} \hat{f}(\boldsymbol{x}_{S_j^\tau \cup \{j\}}, \boldsymbol{x}^{(i)}_{-\{S_j^\tau \cup \{j\}\}}) - \hat{f}(\boldsymbol{x}_{S_j^\tau}, \boldsymbol{x}^{(i)}_{-S_j^\tau})$$

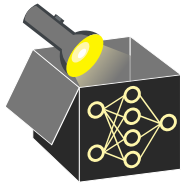
  $\rightsquigarrow$ $\hat{f}_S$ marginalizes over all features not in $S$ using all obs. $i = 1, \ldots, n$
  $\rightsquigarrow$ Exact computation requires $|P|! \cdot n$ marginal contribution terms

# ESTIMATION: A PRACTICAL PROBLEM

- **Exact computation is infeasible for many features:**
  For $|P| = 10$, the number of permutations is $10! \approx 3.6$ million
  $\rightsquigarrow$ Complexity grows factorially with feature count

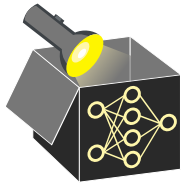


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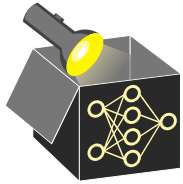
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  Each permut. $\tau$ defines a coal. $S_j^\tau$ needing its own estimate of $\hat{f}_{S_j^\tau}(\boldsymbol{x}_{S_j^\tau})$
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  Instead of computing $|P|! \cdot n$ terms, we approximate using $M$ random samples of permutations $\tau$ and data points

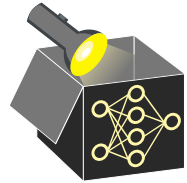
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- **Tradeoff: Accuracy vs. Efficiency**
  Larger $M$ improves Shapley approximation
  $\rightsquigarrow$ Higher cost, but better fidelity to the exact value

# APPROXIMATION ALGORITHM

▸ "Strumbelj et al." 2014

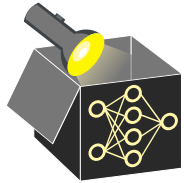


Estimate Shapley value $\phi_j$ of observation $\boldsymbol{x}$ for feature $j$:

- **Input:** $\boldsymbol{x}$ obs. of interest, $j$ feat. of interest, $\hat{f}$ model, $\mathcal{D}$ data, $M$ iterations

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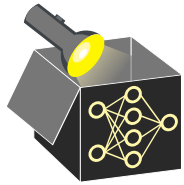
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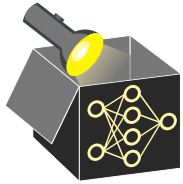
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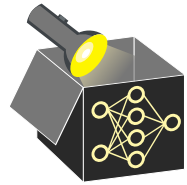
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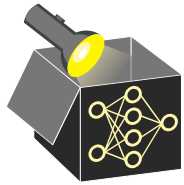
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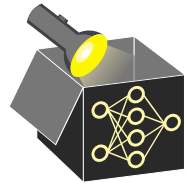
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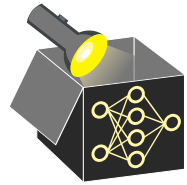
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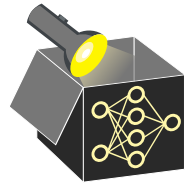
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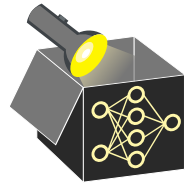
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# APPROXIMATION ALGORITHM

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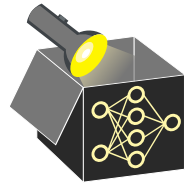
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   5. Compute marginal contribution $\Delta(j, S_m) = \hat{f}(\boldsymbol{x}_{+j}^{(m)}) - \hat{f}(\boldsymbol{x}_{-j}^{(m)})$
2. Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^{M} \Delta(j, S_m)$

# APPROXIMATION ALGORITHM

▸ "Strumbelj et al." 2014



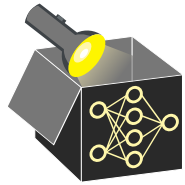
Estimate Shapley value $\phi_j$ of observation $\boldsymbol{x}$ for feature $j$:

- **Input:** $\boldsymbol{x}$ obs. of interest, $j$ feat. of interest, $\hat{f}$ model, $\mathcal{D}$ data, $M$ iterations

1. For $m = 1, \ldots, M$ **do**:
   1. Sample random permut. $\tau = (\tau^{(1)}, \ldots, \tau^{(p)}) \in \Pi$ of feature indices
   2. Let coalition $S_m := S_j^\tau$ be the set of features preceding $j$ in $\tau$
   3. Sample random data point $\mathbf{z}^{(m)} \in \mathcal{D}$ (so-called background data)
   4. Construct two hybrid obs. by combining values from $\boldsymbol{x}$ and $\mathbf{z}^{(m)}$:
      - $\boldsymbol{x}_{+j}^{(m)} = (x_{\tau(1)}, \ldots, x_{\tau(|S_m|)}, x_j, z^{(m)}_{\tau(|S_m|+2)}, \ldots, z^{(m)}_{\tau(p)})$
        $\rightsquigarrow$ includes $\boldsymbol{x}_{S_m \cup \{j\}}$ (features in $S_m \cup \{j\}$ from $\boldsymbol{x}$), rest from $\mathbf{z}^{(m)}$
      - $\boldsymbol{x}_{-j}^{(m)} = (x_{\tau(1)}, \ldots, x_{\tau(|S_m|)}, z_j^{(m)}, z^{(m)}_{\tau(|S_m|+2)}, \ldots, z^{(m)}_{\tau(p)})$
        $\rightsquigarrow$ includes $\boldsymbol{x}_{S_m}$ (features in $S_m$ excl. $x_j$ from $\boldsymbol{x}$), rest from $\mathbf{z}^{(m)}$
   5. Compute marginal contribution $\Delta(j, S_m) = \hat{f}(\boldsymbol{x}_{+j}^{(m)}) - \hat{f}(\boldsymbol{x}_{-j}^{(m)})$
2. Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^{M} \Delta(j, S_m)$

$\rightsquigarrow$ Over $M$ iterations, the PD functions $\hat{f}_{S_m}(\boldsymbol{x}_{S_m})$ and $\hat{f}_{S_m \cup \{j\}}(\boldsymbol{x}_{S_m \cup \{j\}})$ are approximated by $\hat{f}(\boldsymbol{x}_{-j}^{(m)})$ and $\hat{f}(\boldsymbol{x}_{+j}^{(m)})$, where features not in the coalition (to be marginalized) are imputed with vals from random data points $\mathbf{z}^{(m)}$

# SHAPLEY VALUE APPROX. - ILLUSTRATION



**Definition**

$\boldsymbol{x}$: obs. of interest

$\boldsymbol{x}$ with feature values in $\boldsymbol{x}_{S_m}$ (other are replaced)

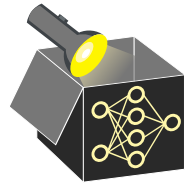
$$\phi_j(\boldsymbol{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\boldsymbol{x}_{+j}{}^{(m)}) - \hat{f}(\boldsymbol{x}_{-j}{}^{(m)})\right]$$

$\boldsymbol{x}$ with feature values in $\boldsymbol{x}_{S_m \cup \{j\}}$

| | Temperature | Humidity | Windspeed | Year |
|---|---|---|---|---|
| $x$ | 10.66 | 56 | 11 | 2012 |
| $x_{+j}$ | 10.66 | 56 | $random: z^{(m)}_{windspeed}$ | 2012 |
| $x_{-j}$ | 10.66 | 56 | $random: z^{(m)}_{windspeed}$ | $random: z^{(m)}_{year}$ |

$j$

# SHAPLEY VALUE APPROX. - ILLUSTRATION



**Definition**

Contribution of feature $j$ to coalition $S_m$

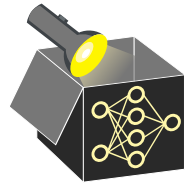
$$\phi_j(\boldsymbol{x}) = \frac{1}{M}\sum_{m=1}^{M}\underbrace{\left[\hat{f}(\boldsymbol{x}_{+j}^{(m)}) - \hat{f}(\boldsymbol{x}_{-j}^{(m)})\right]}_{:=\Delta(j,S_m)}$$

- $\Delta(j, S_m) = \hat{f}(\boldsymbol{x}_{+j}^{(m)}) - \hat{f}(\boldsymbol{x}_{-j}^{(m)})$ is marginal contribution of feature $j$ to coalition $S_m$
- Here: Feature *year* contributes +700 bike rentals if it joins coalition $S_m = \{temp, hum\}$

| | Temperature | Humidity | Windspeed | Year | Count |
|---|---|---|---|---|---|
| $x$ | 10.66 | 56 | 11 | 2012 | 5600 |
| $x_{+j}$ | 10.66 | 56 | $random: z_{windspeed}^{(m)}$ | 2012 | 4900 |
| $x_{-j}$ | 10.66 | 56 | $random: z_{windspeed}^{(m)}$ | $random: z_{year}^{(m)}$ | 700 |

$j$ | $\hat{f}$ | $\Delta(j, S_m)$ marginal contribution

# SHAPLEY VALUE APPROX. - ILLUSTRATION

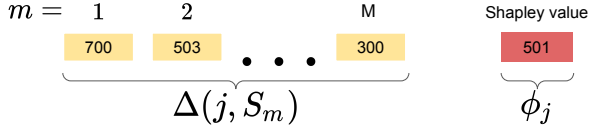


**Definition**

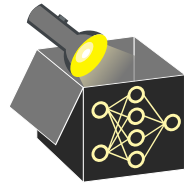
average the contributions of feature *j*

$$\phi_j(\boldsymbol{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\boldsymbol{x}_{+j}^{(m)}) - \hat{f}(\boldsymbol{x}_{-j}^{(m)})\right]$$

- Compute marginal contribution of feature *j* towards the prediction across all randomly drawn feature coalitions $S_1, \ldots, S_m$
- Average all *M* marginal contributions of feature *j*
- Shapley value $\phi_j$ is the payout of feature *j*, i.e., how much feature *year* contributed to the overall prediction in bicycle counts of a specific obs. $\boldsymbol{x}$



# REVISITED: AXIOMS FOR FAIR ATTRIBUTIONS



We adapt the classic Shapley axioms to the setting of model predictions:

- **Efficiency**: Sum of Shapley values adds up to the centered prediction:

$$\sum_{j=1}^{p} \phi_j(\boldsymbol{x}) = \hat{f}(\boldsymbol{x}) - \mathbb{E}_{\boldsymbol{x}}[\hat{f}(\boldsymbol{x})]$$

  $\rightsquigarrow$ All predictive contribution is fully distributed among features

- **Symmetry**: Identical contributors receive equal value:

$$\hat{f}_{S\cup\{j\}}(\boldsymbol{x}_{S\cup\{j\}}) = \hat{f}_{S\cup\{k\}}(\boldsymbol{x}_{S\cup\{k\}}) \; \forall S \subseteq P \setminus \{j,k\} \Rightarrow \phi_j = \phi_k$$

  $\rightsquigarrow$ Interaction effects are shared equitably

- **Dummy (Null Player)**: Irrelevant features receive zero attribution:

$$\hat{f}_{S\cup\{j\}}(\boldsymbol{x}_{S\cup\{j\}}) = \hat{f}_S(\boldsymbol{x}_S) \; \forall S \subseteq P \Rightarrow \phi_j = 0$$

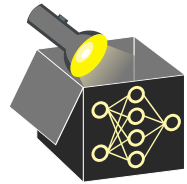
  $\rightsquigarrow$ Shapley value is zero for unused features (e.g., trees or LASSO)

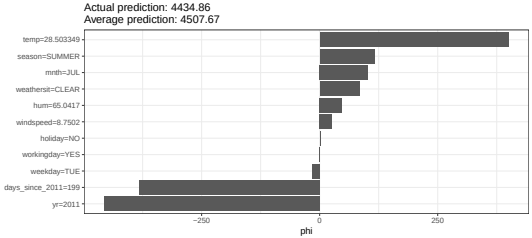
- **Additivity**: Attributions are additive across models:

$$\phi_j(v_1 + v_2) = \phi_j(v_1) + \phi_j(v_2)$$

  $\rightsquigarrow$ Enables combining Shapley values for model ensembles

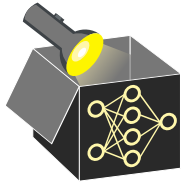
# BIKE SHARING DATASET






- Shapley decomposition for a single prediction in bike sharing dataset
- Model pred.: $\hat{f}(\boldsymbol{x}^{(200)}) =$ **4434.86** vs. dataset avg.: $\mathbb{E}_{\boldsymbol{x}}[\hat{f}(\boldsymbol{x})] =$ **4507.67**
- Total feature attribution: $\sum_j \phi_j =$ **-72.81**
  $\rightsquigarrow$ Explain downward shift from mean prediction
- Temperature (with value 28.5°C) – strongest positive contributor: $+400$
- `yr = 2011` and `days_since_2011 = 199` strongly reduce prediction
  $\rightsquigarrow$ Model captures lower bike demand in 2011 compared to 2012

# ADVANTAGES AND DISADVANTAGES



**Advantages:**

- **Strong theoretical foundation** from cooperative game theory
- **Fair attribution:** Prediction is additively distributed across features
  $\rightsquigarrow$ Easy to interpret for users
- **Contrastive explanations:** Quantify each feature's role in deviating from the average prediction

**Disadvantages:**

- **Comput. cost:** Exact computation scales factorially with feature count
  $\rightsquigarrow$ Without sampling, all $2^p$ coalitions (or $p!$ permuts) must be evaluated
- **Issue with correlated features:** Shapley values may evaluate the model on feature combinations that do not occur in the real data